

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.4 Cross Products

Motivation: We want an operation that will take in two vectors and spit out a vector orthogonal to both vectors.

Definition: If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then the cross product of $\mathbf{a}$ and $\mathbf{b}$ is the vector:
 $\mathbf{a} \times \mathbf{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$.

Remark: The calculation of the cross product is closely related to finding the determinant of a 3 by 3 matrix. So much so, that we often write:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle.$$

Note: Unlike the dot product the result from the cross product is a vector!

Note: While the dot product can be defined in any number of dimensions the cross product is only defined for three dimensional vectors!

Example 1: If $\mathbf{a} = \langle 2, 3, -3 \rangle$ and $\mathbf{b} = \langle 1, -1, 2 \rangle$ then determine each of the following cross products $\mathbf{a} \times \mathbf{b}$ for each of the following:

a) $\mathbf{a} \times \mathbf{b}$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -3 \\ 1 & -1 & 2 \end{vmatrix}$$

b) $\mathbf{b} \times \mathbf{a}$.

c) $\mathbf{a} \times \mathbf{a}$.

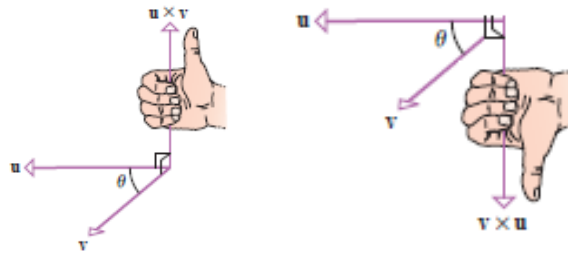
Note: Using the dot product, one can observe that the results in (a) & (b) for this example are orthogonal to both vector $\mathbf{a}$ and vector $\mathbf{b}$.

Theorem: Let $\mathbf{a}$ and $\mathbf{b}$ be vectors in 3-space. Then, $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

Proof: Shown in notes online. You should be able to follow this proof.

A question that we might have is regarding the direction in which $\mathbf{a} \times \mathbf{b}$ points if $\mathbf{a}$ and $\mathbf{b}$ are two non-parallel vectors. While this is possible to determine by investigating the resulting vector

$\mathbf{a} \times \mathbf{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$, the direction can be summarized into what we call the **right-hand rule**.



It is important to note that we do not *prove* this rule here.

Remark: The following identities may remind you of the right hand rule from physics (see page 799 of the book for more details): $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{i} = -\mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$, and $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$.

Theorem: If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta)$.

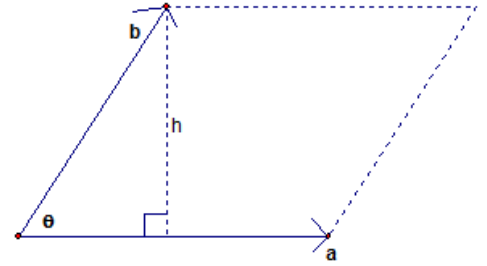
Proof: See page 799 of the text.

Corollary: Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ (in three dimensions) are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

Proof:

Example 2: Suppose that $\mathbf{a}$ and $\mathbf{b}$ are vectors in three dimensions. Find a simple formula for the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$.

A picture of the situation (assuming that the angle is not obtuse) is:



Theorem: The area, A , of a parallelogram that has $\mathbf{a}$ and $\mathbf{b}$ as adjacent sides is given by $A = \|\mathbf{a} \times \mathbf{b}\|$.

Question: How would the formula change if you were asked to find the area of the triangle determined by $\mathbf{a}$ and $\mathbf{b}$?

Theorem: Suppose $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are three dimensional vectors and c is a scalar. Then,

1. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
2. $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (c\mathbf{b})$
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
4. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$
5. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$
6. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Proof: These properties can be proven with typical component wise proofs.

Definition: If $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$, $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, & $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ are vectors in 3-space, then the number

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

is called the scalar triple product of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

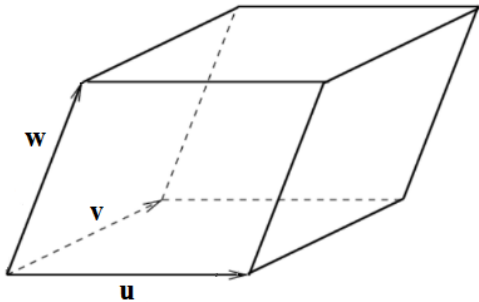
Note: There is no ambiguity to the order of vector operations if this expressed as $\mathbf{u} \cdot \mathbf{v} \times \mathbf{w}$.

Example 3: Let $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - \mathbf{k}$, & $\mathbf{w} = 3\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$. Compute the scalar triple product.

Definition: A parallelepiped is a solid that has 6 quadrilateral faces such that opposing faces are congruent parallelograms.

Example 4: Suppose that you are working with three non-zero vectors, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ which form adjacent sides of a parallelepiped.

- a) What is the volume of the resulting parallelepiped?



- b) Determine the volume of the parallelepiped determined by the vectors in Example 3.

Theorem: Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be nonzero vectors in 3-space.

a) The volume V of the parallelepiped that has $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ as adjacent edges is given by

$$V = |\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$$

b) $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ if and only if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ lie in the same plane.

Proof: a) was completed above and part b) is presented in the notes online.

Example 5: Do the vectors from Example 3 all reside within the same plane?